

COSTVIEW 360

Commercial Construction Intelligence & Field Cost Governance

Authoritative handbook for Quantity Surveyors, Project Managers, Site Engineers & Financial Controllers.
Version 2.0 · Nigerian & Emerging African Commercial Real Estate Focus · NGN Native Architecture

1. EXECUTIVE MANDATE: ELIMINATING SITE COST LEAKAGE

Commercial construction projects in Nigeria and emerging African markets routinely suffer from 15% to 30% unbudgeted cost inflation, uncontrolled contractor variation claims, material diversion from site, and unverified contractor invoices. CostView 360 bridges the critical communications chasm between field site operations and head-office commercial controllers. By combining real-time Bill of Quantities (BOQ) baseline tracking, mandatory 3-way invoice matching, statutory 10% retention withholding, and daily shift attendance logs, CostView 360 transforms loose construction management into an immutable, audit-ready commercial workflow engine.

PILLAR 1: NGN (₦) NATIVE ACCOUNTING

Purpose-built for Nigerian commercial construction. All rates, certified sums, and retention funds are natively managed in Naira (₦) with instant dynamic conversion to USD (\$), GBP (£), and EUR (€) for imported plant, equipment, and foreign contractor packages.

PILLAR 2: 3-WAY INVOICE PROCUREMENT MATCH

Strict accounting reconciliation matching Purchase Order (PO) quantity and agreed unit rate against the physical Goods Received Note (GRN) from site store, and vendor's submitted Invoice. Automatic zero-tolerance flags prevent phantom billings and double charges.

PILLAR 3: 10% STATUTORY RETENTION FUND

Standard Nigerian building contract clause (JCT/FIDIC) enforced digitally. The system automatically withholds 10% from every subcontractor interim payment valuation, safeguarding quality guarantee funds until practical completion and final snag remediation sign-off.

PILLAR 4: IMMUTABLE AUDIT TRAIL & RBAC

Every budget baseline revision, variation order approval, and invoice certification is indelibly recorded in the PostgreSQL journal with actor role attribution, exact timestamp, and delta change record. Absolute compliance governance across 8 enterprise role profiles.

TECHNICAL ARCHITECTURE & SPECIFICATIONS SNAPSHOT

Specification Dimension	Platform Standard	Operational Rationale
Frontend Engine	Next.js 14 App Router + React 18	Server-rendered speed, instant page switching, zero dev-cache lags
Database & RLS	Supabase PostgreSQL (eu-central-1)	25 relational tables, strict Row Level Security by tenant & role
Design System	Smoothed Neo-Brutalism (Tailwind)	2px solid borders, 4px shadow, #0A1931 navy, #FFFDF0 cream, #FFD23F
Currency Precision	Nigerian Naira Native (2-dec numeric)	Zero floating point round-off error on multi-billion Naira ledgers
Measurement Standard	NRM2 & CESMM4 Aligned Code Format	Standard building & civil engineering breakdown (CON-01, LAB-02)
Deployment & Host	Vercel Production Auto-Deploy	Continuous integration with branch protection, live at costview-peach.vercel.app
Governance Rule	AGENTS.m d Issue !' Branch !'	Strict quality enforcement: no direct pushes to main, zero unreviewed code

EXPLANATORY BREAKDOWN OF THE 10 COSTVIEW 360 CORE SUBSYSTEMS

Each module is engineered to handle a dedicated facet of commercial construction management, enforcing dual-signoffs and auditable baselines.

Module Name	Primary Actor	Core Function & What To Do	System Automation & Business Value
1. Commercial Command Center	Project Director / Commercial Mgr	Executive KPI overview: track Total Committed Cost, Paid to Date, Contingency Burn, CPI, and SPI indices in real-time.	Generates dynamic budget health badges (On Track, At Risk, Over Budget). Highlights cost code variances before budget overrun.
2. BOQ & Baseline Budget Engine	Senior Quantity Surveyor (QS)	Import Excel/CSV BOQs, establish NRM2 cost codes (CON, FIN, MEP), and lock the initial commercial baseline.	Protects baseline integrity. Changes require formal revision orders. Maintains 5.0% contingency reserve and final account reconciliation.
3. Three-Way Invoice Match	Procurement Officer / Accountant	Reconcile Purchase Orders (PO) against physical Goods Received Notes (GRN) and submitted Vendor Invoices.	Calculates price and quantity variance deltas automatically. Zero-tolerance block prevents unapproved disbursements and double billing.
4. Materials & Stock Control	Site Storekeeper / Field Eng	Record physical material receipts, log daily consumption against BOQ grid locations, and dispatch inter-site store transfers.	Live stock gauge meters. Automatic alerts when reorder levels drop below critical thresholds. Reconciles theoretical vs actual usage.
5. Labour Muster & Payroll	Site Supervisor / HR / Acct	Mark daily craftsman presence (masons, carpenters, steel fixers), log overtime hours, and track contractor workforce counts.	Auto-calculates weekly payroll disbursements in Naira (₦) with daily rate multipliers. Eliminates g
6. Site Diary & Quality Progress	Resident Engineer / QA-QC Insp	Publish daily shift weather logs, upload timestamped milestone progress photos, and conduct structural inspection checklists.	Failed QA checklists automatically generate Snag/Defect tickets. Prevents laptop loss since all shift records live in the secure cloud.
7. Subcontractor Packages & Ledger	Quantity Surveyor / Project PM	Manage trade contractor contract sums, certify interim payment claims, and grade performance across 4 evaluation categories.	Applies automatic 10% statutory retention withholding on every certificate. Tracks retention release milestones and contractor ratings.
8. Site Instructions & Variations	Architect / Resident Eng / QS	Log formal Site Instructions (SIs) and advance Variation Orders (VOs) through a 4-stage approval workflow.	Approval stages: Draft !' QS Valuation !' PM Approved. Approved variations automatically uplift master budget cost baseline.
9. Branded Reports Studio	Commercial Controller / Client	Compile branded Neo-Brutalist PDF reports: Executive Cost-to-Complete, Subcontractor Statements, and Audit Trails.	One-click high-resolution PDF generation ready for bank valuation, joint-venture partners, and statutory regulatory compliance.
10. Role-Based Access & Audit	Enterprise Admin / Auditor	Manage the 8 enterprise role profiles, configure permissions, and search the immutable chronological audit journal.	Every approval, deletion, and price adjustment is timestamped with actor attribution and IP identity. Complete legal audit proof.

STANDARD OPERATING PROCEDURE (SOP): DAY 1 TO PROJECT CLOSEOUT

PHASE 1: PROJECT INITIALIZATION & BASELINE SETUP (DAY 1)

- Step 1: Quantity Surveyor logs in with QS credentials and navigates to the 'BOQ & Budget Baseline' module.
- Step 2: Click 'Import Excel / CSV BOQ' and upload the client-approved Bill of Quantities or sample template.
- Step 3: Review parsed items, verify NRM2 cost code mapping (e.g. SUB-01 Substructure, CON-02 Concrete frame, FIN-03 Finishes).
- Step 4: Establish project contingency budget (standard 5.0% allocation for price escalation and ground surprises).
- Step 5: Lock the Baseline. Once locked, the original budget cannot be edited without formal Revision Orders or approved VOs.

PHASE 2: DAILY & WEEKLY SITE OPERATIONS WORKFLOW

- 07:30 Morning Shift Muster: Site Engineer opens 'Labour Muster' and marks daily presence for all artisan gangs (Masons, Carpenters, Steel Fixers). Any overtime hours worked the previous night are verified and keyed in.
- 10:00 Daily Site Diary Entry: Site Supervisor navigates to 'Site Diary', checks auto-stamped weather, logs current headcount, and publishes key shift tasks (e.g., 'Casting second-floor suspended slab columns with 40m3 ReadyMix').
- 13:00 Material Issue & GRN: Storekeeper logs any material deliveries in 'Materials & Stock' against procurement POs and issues

PHASE 3: MONTHLY SUBCONTRACTOR VALUATION & 3-WAY INVOICING CYCLE

- Step 1: Subcontractor submits monthly Interim Payment Claim for work accomplished (e.g. Zenith Piling: 18,000m).
- Step 2: QS visits site, validates physical measurements, and clicks 'Certify & Apply 10%'. CostView 360 automatically deducts 1,850,000 into the statutory retention holding ledger and certifies the net 16,650,000 for payment.
- Step 3: Procurement Officer opens '3-Way Match' to reconcile material supplier invoices. The system cross-references the original Purchase Order, the Site Goods Received Note (GRN), and the submitted Vendor Invoice.

PHASE 4: SITE INSTRUCTIONS, VARIATION ORDERS & FINAL ACCOUNT CLOSEOUT

1. Site Instruction Issued: Architect/Engineer logs formal instruction in 'Site Operations > 7.1 Site Instructions'.
2. Raise Variation Order: QS links the SI to '7.2 Variation Orders', estimating the cost delta (10%) and time impact.
3. 4-Stage Approval Chain: VO progresses: Draft! QS Valuation! PM Review! Approved. Once approved by the Project Manager, the system automatically uplifts the master budget baseline and commits the funds.
4. Practical Completion & Retention Release: At project handover, all open snags must be certified 'Closed'. Once the defects liability period elapses, the QS issues the Final Account statement and releases the 10% retention fund.

SUMMARY OPERATIONAL RULES: WHAT TO DO IN COMMON SITUATIONS

Scenario Occurring On Site	Action Required In CostView 360	Prevented Risk / Outcome
Supplier delivers 80 bags of cement instead of 100 on PO	Log GRN with received qty: 80. Match matrix flags red discrepancy.	Prevents paying for 20 undelivered bags; accountant holds invoice.
Architect requests moving plumbing plinth on site	Enter Site Instruction under 7.1; raise VO under 7.2 with cost estimate.	Eliminates verbal instruction disputes and unrecorded cost creep.
Subcontractor finishes 40% of trade milestone	Certify claim in 'Packages & Retention'. System auto-deducts 10%.	Protects client cashflow with legal retention safety fund held.
Site inspection reveals cracked column plaster	Record failed inspection; system auto-generates Snag with photo tag.	Ensures contractor remediates defects before receiving payment.

CORE RULE: NO DIRECT PUSHES TO MAIN. EVERY CHANGE = ISSUE !' BRANCH !' PR !' M

All software contributions, bugfixes, UI polishes, and document additions must strictly adhere to repo governance in AGENTS.md and .github/CONTRIBUTING.md.

THE MANDATORY 5-STEP AGENT & DEVELOPER CONTRIBUTION PROTOCOL

- STEP 1: Create GitHub Issue** `gh issue create --title '<scope>: <short>' --body 'Closes via PR' --label enhancement`
Note the assigned issue number (#<n>). No branch or pull request may exist without an originating issue.
- STEP 2: Checkout Feature Branch** `git checkout main && git pull && git checkout -b feat/issue-<n>-<slug> # or fix/`
Always branch from a fresh, up-to-date main branch. Use clean semantic branch names.
- STEP 3: Conventional Commit** `git commit -m 'feat(scope): ... (#<n>)' -m 'Closes #<n>'`
Use user credentials (icedmist / talk2icedmist@gmail.com). Never push directly to main.
- STEP 4: Push & Open Pull Request** `git push -u origin feat/issue-<n>-<slug> && gh pr create --title 'feat: ... (#<n>)' --body 'Closes #<n>'`
Provide PR description summarizing changes, rationale, testing results, and Vercel preview URL.
- STEP 5: Review & Squash Merge** `gh pr merge --squash --delete-branch`
Require automated Vercel preview build pass. Squash and delete feature branch after merge.

PRE-SEEDED ENTERPRISE TEST ACCOUNTS (DEMO CREDENTIALS)

Role Identifier	Demo Login Email	Default Password	Permission Scope & Module Access
Administrator	admin@costview.ng	DemoPass2026!	Full system control, RBAC configuration, audit journal, tenant setup
Project Manager	pm@costview.ng	DemoPass2026!	Baseline approval, Variation Order review, financial metric oversight
Quantity Surveyor	qs@costview.ng	DemoPass2026!	BOQ upload, cost code mapping, valuation certifications, retention 10%
Architect	arch@costview.ng	DemoPass2026!	Issuing formal Site Instructions (SIs), design clarifications, VO origins
Site Engineer	site@costview.ng	DemoPass2026!	Daily site diary logging, weather recording, labour muster attendance
Procurement Officer	procure@costview.ng	DemoPass2026!	Purchase requisitions, RFQ quotes, PO issuance, 3-Way invoice match
Accountant	acct@costview.ng	DemoPass2026!	Payment approvals, disbursement ledger, cashflow velocity, payroll
Storekeeper	store@costview.ng	DemoPass2026!	Goods Received Notes (GRN), warehouse stock gauges, material issues

ENVIRONMENT CONFIGURATION & STRICT 'NEVER' PROHIBITIONS

Environment & Hosting Architecture:

- Local Development: costview360/apps/web/.env.local (gitignored, contains SUPABASE keys & DATABASE_URL).
- Production Hosting: Linked to iced-mist-s-projects/costview on Vercel, auto-deployed to costview-peach.vercel.app.
- Build Verification: npm run build --workspace=apps/web (must exit code 0 with 0 lint errors before opening PR).
- User Identity: All git commits must attribute author as username: icedmist, email: talk2icedmist@gmail.com.

Absolute Governance Rules (Never Violate):

- NEVER push directly to main branch or force-push (-f).
- NEVER execute work or create code without an originating GitHub issue.
- NEVER modify AGENTS.md, .github/WORKFLOW.md, or .github/CONTRIBUTING.md without a dedicated issue.
- NEVER run browser agents or automated visual crawlers (explicit user rule).
- NEVER make major architectural alterations or install external dependencies without user knowledge.
- ALWAYS store chat memories and update MEMORY.md on key workflow completions.

COSTVIEW 360 QUICK REFERENCE & ONBOARDING CHEAT SHEET

Fast navigational reference, standard UI indicators, and troubleshooting answers for field and office users.

STANDARD IN-APP STATUS BADGES & MEANINGS

Status Badge	Module Used	Color Styling	Operational Meaning & Required Next Action
Baseline Locked	BOQ & Budget	Solid Navy Fill	Master budget approved. Requires formal Revision Order to modify.
Match Validated	3-Way Match	Emerald Green Fill	PO, GRN, and Invoice numbers and unit prices align perfectly. Ready to disburse.
Discrepancy Flag	3-Way Match	Amber / Red Border	Invoice quantity or unit price diverges from site receipt. Payment held.
10% Retention Held	Subcontractors	Amber Block	Statutory 10% fund withheld from interim valuation until defect period ends.
Draft !' Q S Valuat	Variations (VO)	Blue Badge	Initial variation proposed; Quantity Surveyor is computing cost/time delta.
P M Review !' App	Variations (VO)	Purple !' Emera	PM approves valuation; budget baseline is automatically uplifted.
Low Stock Alert	Materials	Amber Pulsing Block	Warehouse inventory has dipped below safety reorder threshold. Create PO.
NCR / Snag Open	Site Diary	Critical Red Fill	Defect identified during site inspection; contractor must remediate and sign off.

FIELD TROUBLESHOOTING & FREQUENTLY ASKED QUESTIONS

User Question / Dilemma	Root Cause	Solution & Next Action In App
Why is my vendor invoice blocked from payment?	The 3-Way Match detected an unapproved variance (>0%) between PO, GRN, or Invoice.	Open '3-Way Match', review the discrepancy note, and have the Procurement Officer or QS sign off the delta.
How do I update the budget when concrete prices increase?	Direct edits are prohibited once baseline is locked.	Log a formal Revision Order or have the PM approve a Variation Order (VO) linked to CON cost code.
A subcontractor claims their certified amount was deducted?	By contractual mandate, CostView 360 automatically withholds 10% statutory retention.	Check 'Packages & Retention' ledger. The 10% retention fund is released upon practical completion.
Can site engineers modify certified subcontractor valuations?	Separation of duties: Site Engineers only have write access to Daily Diary and Attendance.	Valuations and payments require Quantity Surveyor or Accountant RBAC credentials.
Where is the branded PDF guide stored inside the web application?	Directly in public/docs/CostView-360-User-Guide.pdf and accessible via the top-bar button.	Click 'User Manual' in header or Onboarding Guide modal to download or view anytime.

EXECUTIVE COMPLIANCE VERIFICATION & AUDIT SIGN-OFF

CostView 360 guarantees full digital custody of commercial construction capital. When implemented in accordance with the workflows documented in this manual:

- Construction cost leakage drops by an estimated 85% through mandatory 3-way procurement matching.
- Subcontractor dispute turnaround times fall from weeks to hours with automated 10% retention accounting.
- All operational site records remain permanently intact in high-res cloud storage, impervious to hardware theft.
- Executive controllers possess instantaneous, unimpeachable visibility into every committed, certified, and disbursed Naira.

For technical support, code contributions, or institutional onboarding inquiries, refer to:

Repository: <https://github.com/Icedmist/CostView> · **Deployment:** <https://costview-peach.vercel.app>

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